



PROBABILITY THEORY

MTM2592-Gr:2

10TH WEEK

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SPECIAL DISCRETE PROBABILITY DISTRIBUTIONS

1. THE DISCRETE UNIFORM DISTRIBUTION

If the random variable X takes the values $x_1, x_2, \dots, x_n$ with equal probabilities, then the discrete uniform distribution is given by

$$f(x; n) = \frac{1}{n}; \quad x = x_1, x_2, \dots, x_n$$

2. THE BERNOULLI DISTRIBUTION

Let an experiment consist of repeated trials and each trial be independent. If each trial **has two possible outcomes**, i.e. Success or failure,

X: Number of successes in a Bernoulli experiment;

$$f(x; p) = p^x (1-p)^{1-x} \quad x = 0, 1$$

3. BINOMIAL DISTRIBUTION (TWO TERM DISTRIBUTION)

Let **a Bernoulli trial be repeated n times** and the probability of an outcome be p .

X: Number of successes in n Bernoulli experiments

$$b(x; n, p) = \binom{n}{x} p^x (1-p)^{n-x} \quad x = 0, 1, 2, \dots, n$$

4. MULTINOMIAL DISTRIBUTION

There are n independent experiments with only two possible outcomes in the binomial distribution. Consider the fact that an experiment has more than two outcomes. If this experiment is repeated n times, then a Multinomial distribution is used. This distribution is also called Generalized Binomial Distribution.

x_1 represents the number of occurrence of A_1 ,

x_2 represents the number of occurrence of A_2 ,

.....

x_k represents the number of occurrence of A_k

then the probability

$$P(X_1 = x_1, X_2 = x_2, \dots, X_k = x_k)$$
$$f(x_1, x_2, \dots, x_k; n, p_1, p_2, \dots, p_k) = \binom{n}{x_1, x_2, \dots, x_k} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$$

is called Multinomial distribution.

5. GEOMETRIC DISTRIBUTION

Consider a Bernoulli trial. If X represents the number of trials on which the first success occurs, then the geometric distribution is given by

$$g(x; p) = q^{x-1} p \quad x = 1, 2, 3, \dots$$

X: Number of Bernoulli trials required to achieve a success

6. NEGATIVE BINOMIAL (PASCAL) DISTRIBUTION

Consider a Bernoulli trial. If X represents the number of trials on which the **k-th success occurs**, then the negative binomial distribution is given by

$$b^*(x; k, p) = \binom{x-1}{k-1} p^k q^{x-k} \quad x = k, k+1, k+2, \dots$$

which means $P(X=x)$. It is also called Pascal probability function.

X: Number of Bernoulli trials required to achieve k success

7. HYPERGEOMETRIC DISTRIBUTION

Definition: Assume that a set of N elements consists of two type of items, namely M type A, and $N-M$ type B. Let n items are randomly selected from the set **without replacing them back**.

If X represents the number of chosen items with the type A, the distribution

$$h(x; n, N, M) = \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}} \quad x = 0, 1, 2, \dots, n_k \quad n_k = \begin{cases} n, & n \leq M \\ M, & n > M \end{cases}$$

is called hypergeometric distribution.

In this function, the probability function has three parameters such as N , M and n . In order to use the hypergeometric distribution, the result of one of the experiments must affect other experiments. In other words, successive experiments are dependent.

If the items are selected with replacing them back, then problem fits the binomial distribution.

Let's show that the hypergeometric distribution is a probability function.

i) $h(x; n, N, M) \geq 0$

ii) By using the theorem

$$\sum_{x=0}^n \binom{M}{x} \binom{N-M}{n-x} = \binom{N}{n}$$

$$\sum_{x=0}^n h(x; n, N, M) = \sum_{x=0}^n \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}} = \frac{1}{\binom{N}{n}} \sum_{x=0}^n \binom{M}{x} \binom{N-M}{n-x} = \frac{1}{\binom{N}{n}} \binom{N}{n} = 1$$

Expected Value and Variance Of The Hypergeometric Distribution

By using the definition of the expected value and variance of an X random variable,

$$\mu = E(X) = \sum_{x=0}^n x \cdot \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}$$

$$= \frac{1}{\binom{N}{n}} \sum_{x=0}^n \frac{x \cdot M!}{x!(M-x)!} \binom{N-M}{n-x} = \frac{1}{\binom{N}{n}} \sum_{x=1}^n \frac{x \cdot M!}{x!(M-x)!} \binom{N-M}{n-x} = \frac{1}{\binom{N}{n}} \sum_{x=1}^n \frac{M!}{(x-1)!(M-x)!} \binom{N-M}{n-x}$$

$$= \frac{M}{\binom{N}{n}} \sum_{x=1}^n \frac{(M-1)!}{(x-1)!(M-x)!} \binom{N-M}{n-x} = \frac{M}{\binom{N}{n}} \sum_{y=0}^{n-1} \frac{(M-1)!}{y!(M-1-y)!} \binom{N-M}{n-1-y}$$

$y=x-1$ $m=n-1$

$$= \frac{M}{\binom{N}{n}} \sum_{y=0}^m \binom{M-1}{y} \binom{N-M}{m-y} = \frac{M}{\binom{N}{n}} \binom{N-1}{m} = \frac{M}{\binom{N}{n}} \binom{N-1}{n-1} = M \frac{\binom{N-1}{n-1} n!}{(n-1)!(N-n)!} = \frac{Mn}{N}$$

N

Also $M / N = p$

$$E(X) = \mu = \frac{M}{N} \cdot n = p \cdot n$$

This expected value is the same as the expected value of the Binomial distribution. It is not possible to say the same for variance.

$$E(X^2) = \sum_{x=0}^n x^2 \cdot \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}$$

Instead of x^2 , $x(x-1)+x$ is written.

$$= \sum_{x=0}^n [x(x-1) + x] \frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}} = \frac{M(M-1)}{\binom{N}{n}} \sum_{x=0}^n \frac{x(x-1)(M-2)!}{(M-x)!x!} \binom{N-M}{n-x} + E(X)$$

$$= \frac{M(M-1)}{\binom{N}{n}} \sum_{x=2}^n \frac{(M-2)!}{(M-x)!(x-2)!} \binom{N-M}{n-x} + E(X)$$

$$= \frac{M(M-1)}{\binom{N}{n}} \sum_{x=2}^n \frac{(M-2)!}{(M-x)!(x-2)!} \binom{N-M}{n-x} + E(X)$$

$$y=x-2$$

$$n-2=m$$

$$= \frac{M(M-1)}{\binom{N}{n}} \sum_{y=0}^m \frac{(M-2)!}{(M-2-y)!y!} \binom{N-M}{m-y} + E(X) = \frac{M(M-1)}{\binom{N}{n}} \sum_{y=0}^m \binom{M-2}{y} \binom{N-M}{m-y} + E(X)$$

$$= \frac{M(M-1)}{\binom{N}{n}} \binom{N-2}{m} + E(X) = \frac{M(M-1)}{\binom{N}{n}} \binom{N-2}{n-2} + E(X)$$

$$= \frac{M(M-1)n(n-1)}{N(N-1)} + \frac{Mn}{N}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= \frac{nM(N-M)(N-n)}{N^2(N-1)} = \frac{N-n}{N-1} \cdot n \cdot \frac{M}{N} \left(1 - \frac{M}{N}\right)$$

$$\frac{M}{N} = p \quad \text{ve} \quad 1 - \frac{M}{N} = q$$

$$\text{Var}(X) = npq \frac{N-n}{N-1}$$

While the arithmetic means of the binomial distribution and the hypergeometric distribution are the same, the variances are different. The $(N-n) / (N-1)$ multiplier in the variance of the hypergeometric distribution creates this difference. This factor is called the **correction factor**.

If $N \rightarrow \infty$ this multiplier is equal to 1. Thus, the variance of the hypergeometric distribution is equal to the npq , just like the variance of the Binomial distribution.

Example: A box consists of 4 White and 11 black balls. If 3 balls are selected without replacing them back, what is the probability that

a) 2 of them is white,

b) At least one of them is white?

Solution:

$$a) P(X = 2) = h(X = 2; n = 3, N = 15, M = 4) = \frac{\binom{4}{2} \binom{11}{1}}{\binom{15}{3}} = 0.1451$$

$$b) P(X \geq 1) = 1 - P(X = 0) = 1 - h(X = 0; n = 3, N = 15, M = 4) = 1 - \frac{\binom{4}{0} \binom{11}{3}}{\binom{15}{3}} = 0.6374$$

Example: There are 3 defective and 7 non-defective items in a box. 3 items are selected without replacing them back. Find the probability function of the number of the defective items.

Solution: $N = 10, M = 3, n = 3$

$$h(X; n = 3, N = 10, M = 3) = \frac{\binom{3}{x} \binom{7}{3-x}}{\binom{10}{3}} \quad x = 0, 1, 2, 3$$

X	0	1	2	3
f(X)	35/120	63/120	21/120	1/120

8. MULTIVARIATE HYPERGEOMETRIC DISTRIBUTION

Just as the hypergeometric distribution takes the place of the binomial distribution for sampling without replacement, there also exists a multivariate distribution analogous to the multinomial distribution that applies to sampling without replacement. To derive its formula, let us consider a set of N elements, of which M_1 are elements of the first kind, M_2 are elements of the second kind, $\dots$, and M_k are elements of the k th kind, such that $\sum_{i=1}^k M_i = N$. As in connection with the multinomial distribution, we are interested in the probability of getting x_1 elements (outcomes) of the first kind, x_2 elements of the second kind, $\dots$, and x_k elements of the k th kind, but now we are choosing, without replacement, n of the N elements of the set.

There are $\binom{M_1}{x_1}$ ways of choosing x_1 of the M_1 elements of the first kind, $\binom{M_2}{x_2}$ ways of choosing x_2 of the M_2 elements of the second kind, $\dots$, and $\binom{M_k}{x_k}$ ways of choosing x_k of the M_k elements of the k th kind, and, hence, $\binom{M_1}{x_1} \binom{M_2}{x_2} \dots \binom{M_k}{x_k}$ ways of choosing the required $\sum_{i=1}^k x_i = n$ elements. Since there are $\binom{N}{n}$ ways of choosing n of the N elements in the set and we assume that they are all equally likely (which is what we mean when we say that the selection is random), it follows that the desired probability is given by $\binom{M_1}{x_1} \binom{M_2}{x_2} \dots \binom{M_k}{x_k} / \binom{N}{n}$.

Definition: The random variables $X_1, X_2, \dots, X_k$ have a multivariate hypergeometric distribution and they are referred to as multivariate hypergeometric random variables if and only if their joint probability distribution is given by

$$f(x_1, x_2, \dots, x_k; n, M_1, M_2, \dots, M_k) = \frac{\binom{M_1}{x_1} \cdot \binom{M_2}{x_2} \cdots \binom{M_k}{x_k}}{\binom{N}{n}}$$

for $x_i = 0, 1, \dots, n_i$ and $x_i \leq M_i$ for each i , where

$$\sum_{i=1}^k x_i = n, \quad \sum_{i=1}^k M_i = N$$

Thus, the joint distribution of the random variables under consideration, that is, the distribution of the numbers of outcomes of the different kinds, is a multivariate hypergeometric distribution with the parameters $n, M_1, M_2, \dots$, and M_k .

EXAMPLE

A panel of prospective jurors includes six married men, three single men, seven married women, and four single women. If the selection is random, what is the probability that a jury will consist of four married men, one single man, five married women, and two single women?

Solution

Substituting $x_1 = 4$, $x_2 = 1$, $x_3 = 5$, $x_4 = 2$, $M_1 = 6$, $M_2 = 3$, $M_3 = 7$, $M_4 = 4$, $N = 20$, and $n = 12$ into the formula of Definition , we get

$$\begin{aligned} f(4, 1, 5, 2; 12, 6, 3, 7, 4) &= \frac{\binom{6}{4} \binom{3}{1} \binom{7}{5} \binom{4}{2}}{\binom{20}{12}} \\ &= 0.0450 \end{aligned}$$

9. THE POISSON DISTRIBUTION

When n is large, the calculation of binomial probabilities gets harder. In this case, we will investigate limiting form of the binomial distribution when $n \rightarrow \infty, p \rightarrow 0$, while np remains constant. Letting this constant be λ .

$$\lambda = np \rightarrow p = \lambda/n$$

$$b(x; n, p) = \binom{n}{x} p^x (1-p)^{n-x} \quad x = 0, 1, 2, \dots, n$$

$$b(x; n, p) = \binom{n}{x} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x}$$

$$= \frac{n(n-1)(n-2) \cdot \dots \cdot (n-x+1)}{x!} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x}$$

$$= \frac{n(n-1) \dots (n-x+1)}{n \cdot n \cdot \dots \cdot n} \left(\frac{\lambda^x}{x!}\right) \left(1 - \frac{\lambda}{n}\right)^{n-x}$$

$$= \underbrace{\left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{x-1}{n}\right)}_{\rightarrow 1} \underbrace{\left(\frac{\lambda^x}{x!}\right) \left(1 - \frac{\lambda}{n}\right)^n}_{\rightarrow \frac{\lambda^x}{x!} e^{-\lambda}} \underbrace{\left(1 - \frac{\lambda}{n}\right)^{-x}}_{\rightarrow 1} = \frac{\lambda^x e^{-\lambda}}{x!}$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \rightarrow e$$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n \rightarrow e^{-1}$$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{x-1}{n}\right) = 1$$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n \rightarrow e^{-\lambda}$$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^{-x} = 1$$

Poisson Approach to Binomial Distribution

In the limit when $n \rightarrow \infty$ $p \rightarrow 0$ and λ remains constant, the number of successes is a random variable having a Poisson distribution with the parameter λ .

In general, the Poisson distribution will provide a good approximation to binomial probabilities when

$$n \geq 20, \quad p \leq 0.05$$

when

$$n \geq 100, \quad np \leq 10$$

the approximation will generally be excellent.

$$np = \lambda \quad p = \frac{\lambda}{n}$$

$$\lim_{n \rightarrow \infty} \binom{n}{x} p^x q^{n-x} \rightarrow \frac{\lambda^x e^{-\lambda}}{x!}$$

$$\lim_{n \rightarrow \infty} b(X; n, p) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad \lambda = np$$

Definition: A random variable has a Poisson distribution and it is referred to as a Poisson random variable if and only if its probability distribution is given by

$$p(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} \quad \text{for } x = 0, 1, 2, \dots$$

This distribution has one parameter. This parameter is λ

λ shows the average number of events!

$$\lambda = np \qquad e^{\lambda} = 1 + \lambda + \frac{\lambda^2}{2!} + \dots + \frac{\lambda^x}{x!} + \dots$$

i) $p(x; \lambda) \geq 0$

ii)
$$p(x; \lambda) = \sum_{x=0}^{\infty} \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = e^{-\lambda} \left(1 + \frac{\lambda}{1} + \frac{\lambda^2}{2!} + \dots + \frac{\lambda^n}{n!} + \dots \right) = e^{-\lambda} e^{\lambda} = 1$$

Moment Generating Function, Expected Value and Variance Of The Poisson Distribution

a) By using the definition of moment generating function;

$$\begin{aligned} M(t) = E[e^{tX}] &= \sum_{x=0}^{\infty} e^{tx} p(x; \lambda) = \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda} \lambda^x}{x!} \\ &= \sum_{x=0}^{\infty} \frac{(e^t \lambda)^x e^{-\lambda}}{x!} \end{aligned}$$

By taking $e^t \lambda = a$

$$= \sum_{x=0}^{\infty} \frac{a^x e^{-\lambda}}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{a^x}{x!}$$

In the last equation, e^a is written instead of the total expression.

$$M(t) = e^{-\lambda} \cdot e^{e^t \lambda} = e^{(e^t - 1)\lambda}$$

$$M'(0) = E(X) = \mu = \lambda$$

$$M''(0) = \lambda(1 + \lambda)$$

$$\text{Var}(X) = M''(0) - [M'(0)]^2$$

$$= \lambda(1 + \lambda) - \lambda^2$$

$$\text{Var}(X) = \lambda$$

The arithmetic mean and variance of the Poisson distribution are the same and equal λ . ($\lambda = np$)

b) By using the definition of the expected value and variance of a X random variable,

$$E(X) = \sum_{x=0}^{\infty} x \cdot p(x; \lambda) = \sum_{x=0}^{\infty} x \cdot \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \lambda \sum_{x=1}^{\infty} \frac{\lambda^{(x-1)}}{(x-1)!} = e^{-\lambda} \lambda \underbrace{\sum_{t=0}^{\infty} \frac{\lambda^t}{t!}}_{e^{\lambda}} = e^{-\lambda} \lambda e^{\lambda} = \lambda$$

$t=x-1$

$$E(X) = \lambda$$

$$\begin{aligned} E(X^2) &= \sum_{x=0}^{\infty} x^2 \cdot p(x; \lambda) = \sum_{x=0}^{\infty} x^2 \cdot \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{x=1}^{\infty} x^2 \frac{\lambda^x}{x!} \\ &= e^{-\lambda} \sum_{x=1}^{\infty} \lambda x \frac{\lambda^{x-1}}{(x-1)!} = e^{-\lambda} \lambda \sum_{x=1}^{\infty} [(x-1) + 1] \frac{\lambda^{x-1}}{(x-1)!} \\ &= e^{-\lambda} \lambda \sum_{x=2}^{\infty} (x-1) \frac{\lambda^{x-1}}{(x-1)!} + e^{-\lambda} \lambda \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} \\ \leftarrow t=x-2 \quad &= e^{-\lambda} \lambda \sum_{x=2}^{\infty} \frac{\lambda^{x-1}}{(x-2)!} + e^{-\lambda} \lambda \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} \quad t=x-1 \rightarrow \\ &= e^{-\lambda} \lambda^2 e^{\lambda} + e^{-\lambda} \lambda e^{\lambda} \\ &= \lambda^2 + \lambda \end{aligned}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

EXAMPLE

If 2 percent of the books bound at a certain bindery have defective bindings, use the Poisson approximation to the binomial distribution to determine the probability that 5 of 400 books bound by this bindery will have defective bindings.

Solution

Substituting $x = 5$, $\lambda = 400(0.02) = 8$, and $e^{-8} = 0.00034$ into the formula of Definition we get

$$p(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} \Rightarrow p(5; 8) = \frac{8^5 \cdot e^{-8}}{5!} = \frac{(32,768)(0.00034)}{120} = 0.093$$

EXAMPLE

Records show that the probability is 0.00005 that a car will have a flat tire while crossing a certain bridge. Use the Poisson distribution to approximate the binomial probabilities that, among 10,000 cars crossing this bridge,

- (a) exactly two will have a flat tire;
- (b) at most two will have a flat tire.

Solution

- (a) for $x = 2$ and

$\lambda = 10,000(0.00005) = 0.5$, the Poisson probability is $\Rightarrow p(2; 0.5) = \frac{e^{-0.5} (0.5)^2}{2!} = 0.0758$

- (b) for $x = 0, 1$, and 2,

and $\lambda = 0.5$, the Poisson probabilities are 0.6065, 0.3033, and 0.0758. Thus, the

probability that at most 2 of 10,000 cars crossing the bridge will have a flat tire is

$$0.6065 + 0.3033 + 0.0758 = 0.9856$$

EXAMPLE

The average number of trucks arriving on any one day at a truck depot in a certain city is known to be 12. What is the probability that on a given day fewer than 9 trucks will arrive at this depot?

Solution

Let X be the number of trucks arriving on a given day. Then, using $\lambda = 12$, we get

$$P(X < 9) = \sum_{x=0}^8 p(x; 12) = 0.1550$$

EXAMPLE

A certain kind of sheet metal has, on the average, five defects per 10 square feet. If we assume a Poisson distribution, what is the probability that a 15-square-foot sheet of the metal will have at least six defects?

Solution

Let X denote the number of defects in a 15-square-foot sheet of the metal. Then, since the unit of area is 10 square feet, we have

$$\lambda = \alpha t = (5)(1.5) = 7.5$$

and

$$P(X \geq 6) = 1 - P(X \leq 5) = 1 - 0.2414 = 0.7586$$

Example: Let a pair of die be rolled 7 times. Find the probability that 4 times, the sum of the pair comes up 8, by using the Binomial and Poisson distributions.

Solution: $n = 7, x = 4$

$$S = \{(1,1), (1,2), \dots, (6,6)\}$$

$$A = \{(2,6), (3,5), (4,4), (5,3), (6,2)\}$$

$$P(A) = p = \frac{5}{36}$$

$$b(x; n, p) = b(4; 7; 5/36) = \binom{7}{4} \left(\frac{5}{36}\right)^4 \left(\frac{31}{36}\right)^3 = 0.008$$

$$\lambda = np = 7 \cdot \frac{5}{36} = \frac{35}{36}$$

$$p(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-35/36} \left(\frac{35}{36}\right)^4}{4!} = 0.013$$

Example: It is known that 8% of the products are defective in a factory. If 15 products are randomly selected, find the probability that exactly 3 chosen products are defective by using

- a) Poisson Distribution
- b) Binomial Distribution

Solution: $n = 15$, $x = 3$, $p = 0.08$

$$\lambda = np = 15 \times 0.08 = 1,2$$

$$\text{a) } p(x; 1, 2) = \frac{e^{-\lambda} (1, 2)^x}{x!} = \frac{e^{-1,2} (1, 2)^3}{3!} \cong 0,086$$

$$\text{b) } b(3; 15, 0.08) = \binom{15}{3} (0.08)^3 (0.92)^{12} \cong 0,086$$

Example: A certain kind of a cable has on the average, 2 defects per 5 meters. If we assume it fits the Poisson distribution, find the probability that a 12 meter cable has

- a) 3 defects,
- b) At least one defect.

Solution: For the continuous case, we can consider

$$p = \frac{2 \text{ defects}}{5 \text{ meters}} \quad \text{or} \quad \frac{2/5 \text{ defects}}{1 \text{ meters}}$$

$$\lambda = np = 12 \times \frac{2}{5} = 4,8$$

$$p(x; \lambda) = \frac{e^{-4,8} (4,8)^x}{x!}$$

$$\text{a) } p(3; 4.8) = \frac{e^{-4,8} (4,8)^3}{3!} = 0.1517$$

$$\text{b) } P(X \geq 1) = 1 - P(X < 1) = 1 - P(X = 0) = 1 - \frac{e^{-4,8} (4,8)^0}{0!} = 0.9918$$

Example: It has been determined that the escalators in a shopping center fail for an average of 3 days a year.

- a) What is the probability that the escalators will not fail in any given month?
- b) What is the probability that the escalators will fail less than 2 days in any month?
- c) What is the probability that the escalators will not fail next year?

Solution:

$$\lambda = 30 \cdot \frac{3}{365} \cong 0.247$$

$$p(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$\text{a) } p(X = 0; 0.247) = \frac{e^{-0.274} (0.247)^0}{0!} = 0.78$$

$$\text{b) } P(X < 2) = P(X = 0) + P(X = 1) = \frac{e^{-0.274} (0.247)^0}{0!} + \frac{e^{-0.274} (0.247)^1}{1!} = 0.97$$

$$\text{c) } \lambda = 3 \Rightarrow p(X = 0; 3) = \frac{e^{-3} (3)^0}{0!} = 0.049$$

EXAMPLES

1) A balanced coin is tossed 12 times. Find the probability that the coin shows tail 5 times and head 7 times.

$$b(5; 12, 1/2) = \binom{12}{5} (1/2)^5 (1/2)^7$$

2) A candidate applies to get a driver license. If the probability of passing the exam is 0.75, what is the probability that the candidate passes the exam in the 4th application?

$$g(4; 0.75) = (0.25)^3 0.75$$

3) Let X represents the number of born children. Construct a function that calculates how many children should be born in order that r boys are born.

$$b^*(x; r, 0.5) = \binom{x-1}{r-1} (0.5)^r (0.5)^{x-r}$$

By using the constructed probability function, find the probability that The 4th child of a family is their 1st boy,

$$b^*(4; 1, 0.5) = \binom{4-1}{1-1} (0.5)^1 (0.5)^3$$

The 7th child of a family is their 2nd girl,

$$b^*(7; 2, 0.5) = \binom{7-1}{2-1} (0.5)^2 (0.5)^5$$

The 10th child of a family is their 4th or 5th boy?

$$\begin{aligned} & b^*(10; 4, 0.5) + b^*(10; 5, 0.5) \\ &= \binom{10-1}{4-1} (0.5)^4 (0.5)^6 + \binom{10-1}{5-1} (0.5)^5 (0.5)^5 \end{aligned}$$

4) 120 applicants have applied for a job, but only 80 of them have appropriate qualifications? If 5 applicants are randomly selected, find the probability that 2 of them have appropriate qualifications by using The hypergeometric distributions for the exact solution.

$$h(2; 5, 120, 80) = \frac{\binom{80}{2} \binom{40}{3}}{\binom{120}{5}}$$

the binomial distributions for the approximate solution.

$$b(2; 5, 80/120) = \binom{5}{2} (80/120)^2 (40/120)^3$$

5) A bookbinder binds books and it is known that 2% of these books are defective? Find the approximate probability that 5 books are defective in 400 books that have been made by this bookbinder?

$$p(5; 8) = \frac{e^{-8} 8^5}{5!}$$

6) The records show that the probability of tyres' blowing out on a specific bridge is 0.00005. If 10000 cars pass on this bridge in a day, find the probability that exactly 2 cars' tyre will blow-out.

$$p(2; 0.5) = \frac{e^{-0.5} 0.5^2}{2!}$$

at most 2 cars' tyre will blow-out.

$$p(0; 0.5) + p(1; 0.5) + p(2; 0.5) = \frac{e^{-0.5} 0.5^0}{0!} + \frac{e^{-0.5} 0.5^1}{1!} + \frac{e^{-0.5} 0.5^2}{2!}$$

7) There are 3 TV stations in a city. On Saturday evenings, the Channel 1, Channel 2 and Channel 3 have respectively 50%, 30% and 20% of the TV ratings? If 8 people are randomly selected from the tele viewers that watch TV on Saturday evenings in this city, what is the probability that 5 of them watch the Channel 1, 2 of them watch the Channel 2 and 1 of them watch the Channel 3?

$$f(5, 2, 1; 8, 0.5, 0.3, 0.2) = \frac{8!}{5!2!1!} 0.5^5 0.3^2 0.2^1$$

8) There are 6 married and 3 single men, and also 7 married and 4 single women in a community? If some of them are selected in order to form a jury, what is the probability that this jury consists of 4 married and 1 single men, and also 5 married and 2 single women?

$$f(4, 1, 5, 2; 20, 6, 3, 7, 4) = \frac{\binom{6}{4} \binom{3}{1} \binom{7}{5} \binom{4}{2}}{\binom{20}{12}}$$

9) A pair of dies is rolled 4 times. Find the probability that the pair comes up totally 8 two times by using the Poisson and Binomial distributions?

$$p(2; 5/9) = \frac{e^{-5/9} (5/9)^2}{2!}, \quad b(2; 4, 5/36) = \binom{4}{2} (5/36)^2 (31/36)^2$$

10) It is known that 3% of the TV tubes which are produces in a factory are defective. If 120 TV tubes are randomly selected, find the probability that 4 of them are defective by using the Poisson and Binomial distributions.

$$p(4; 3.6) = \frac{e^{-3.6} (3.6)^4}{4!}, \quad b(4; 120, 0.03) = \binom{120}{4} (0.03)^4 (0.97)^{116}$$

11) A box contains 3 defective and 7 non-defective pieces. If 3 pieces are randomly selected without replacing them back, find the probability function of the numbers of defective pieces.

$$h(x; 3, 10, 3) = \frac{\binom{3}{x} \binom{7}{3-x}}{\binom{10}{3}}, x = 0, 1, 2, 3$$

12) 3 applicants from 3 different occupations apply for 3 positions from 3 different cities. If the probability that each applicant is selected for the position in the city he/she stays is $1/3$, then find the probability that at least one of them is selected for the position in the city he/she stays.

$$P(X \geq 1) = 1 - P(X = 0) = 1 - b\left(0; 3, \frac{1}{3}\right) = 1 - \binom{3}{0} \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^3$$

13) Let a die be rolled 12 times. What is the probability that the dies come up 1 two times, 2 three times, 3 one time, 5 three times and 6 one time?

$$\begin{aligned} f(2, 3, 1, 2, 3, 1; 12, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}) \\ = \frac{12!}{2! 3! 1! 2! 3! 1!} \left(\frac{1}{6}\right)^2 \left(\frac{1}{6}\right)^3 \left(\frac{1}{6}\right)^1 \left(\frac{1}{6}\right)^2 \left(\frac{1}{6}\right)^3 \left(\frac{1}{6}\right)^1 \end{aligned}$$

14) Let a die be rolled successively. What is the probability that the die comes up 6 at the third time in the 7th trial.

$$b^*(7; 3, 1/6) = \binom{6}{2} (1/6)^3 (5/6)^4$$

15) 5 cards are selected from a deck with 52 cards. Find the probability function of the number of the chosen aces.

$$h(x; 5, 52, 4) = \frac{\binom{4}{x} \binom{48}{5-x}}{\binom{52}{5}}, x = 0, 1, 2, 3, 4$$

16) 200 syntax errors are randomly distributed into a book with 200 pages. If a random page is selected, what is the probability that this page contains two syntax errors,

$$p(2; 1) = \frac{e^{-1}}{2!}$$

less than two syntax errors.

$$p(0; 1) + p(1; 1) = 2e^{-1}$$

17) Suppose that, averagely, 4 phone connection are made per a minute in a station. Find the probability that exactly 6 phone connection have been made in 2 minutes,

$$p(6; 8) = \frac{e^{-8} 8^6}{6!}$$

at least 3 phone connection have been made in 3 minutes.

$$\begin{aligned} P(X \geq 3) &= 1 - p(0; 12) - p(1; 12) - p(2; 12) = \\ &= 1 - \frac{e^{-12} 12^0}{0!} - \frac{e^{-12} 12^1}{1!} - \frac{e^{-12} 12^2}{2!} \end{aligned}$$

18) A test exam consists of 8 questions and for each question 3 possible answers (only one of them is correct) are given, namely **A**, **B**, **C**. By rolling a die, a student will select **A** if the die comes up 1 or 2, **B** if the die comes up 3 or 4 and **C** if the die comes up 5 or 6. Let X represents the number of correct answers that the student chooses.

Determine the probability function of X .

$$b(x; 8, 1/3) = \binom{8}{x} (1/3)^x (2/3)^{8-x}, x = 0, 1, 2, \dots, 8$$

Find the probability that the student will choose 4 correct answers.

$$f(4; 8, 1/3) = \binom{8}{4} (1/3)^4 (2/3)^4$$

If the student takes this exam so many times, what is the expectation average of the correct answers that he/she has?

$$E(X) = 8/3$$

19) The probability that an eagle hunts a rabbit throughout the day is 0.10. If the daily huntings are assumed to be independent. Determine the probability function of the random variable that represents the number of passing days for the first successful hunting.

$$g(x; 0.1) = 0.1 (0.9)^{x-1}, x = 0, 1, 2, \dots$$

Find the probability that the eagle hunts a rabbit on Thursday if it starts to hunt on Monday.

$$g(4; 0.1) = 0.1 (0.9)^3$$

What is the expected average of the number of days for the first successful hunting?

$$E(X) = \frac{1}{p} = 10$$

20) If X represents the number of success for the Bernoulli trials repeated n times. Determine the probability function of X .

$$b(x; n, p) = \binom{n}{x} p^x (1 - p)^{n-x}, x = 0, 1, 2, \dots, n$$

What should be the number of trials in order to decrease the standard deviation to its half?

$$n/4$$

21) The probability that a tax declaration is filled correctly is 0.60, that contains errors on behalf of an obligator is 0.20, that contains errors on behalf of the government is 0.10 and that contains errors on behalf of both the obligator and the government is 0.10. If 12 tax declaration are randomly selected for inspection, find the probability that five of them are filled correctly, four of them contain errors on behalf of an obligator, two of them contain errors on behalf of the government and one of them contains errors on behalf of both an obligator and the government. (Solve the problem by constructing the probability function.).

$$f(5, 4, 2, 1; 12, 0.6, 0.2, 0.1, 0.1) = \frac{12!}{5! 4! 2! 1!} (0.6)^5 (0.2)^4 (0.1)^2 (0.1)^1$$

22) It is known that 8% of the product are defected in a factory. If 15 products are randomly selected, find the probability that exactly 3 chosen products are defected by using

The Poisson distribution,

$$p(x; 1.2) = \frac{e^{-1.2} 1.2^3}{3!}$$

The Binomial distribution,

$$b(3; 15, 0.08) = \binom{15}{3} (0.08)^3 (0.92)^{12}$$